torch.kaiser_window#

https://pytorch.org/docs/stable/generated/torch.kaiser_window.html#torch.kai

Description:

Computes the Kaiser window with window length window_length and shape parameter beta. Let I_0 be the zeroth order modified Bessel function of the first kind (see torch.i0()) and N = L - 1 if periodic is False and L if periodic is True, where L is the window_length. This function computes: Calling torch.kaiser_window(L, B, periodic=True) is equivalent to calling torch.kaiser_window(L + 1, B, periodic=False)[:-1]). The periodic argument is intended as a helpful shorthand to produce a periodic window as input to functions like torch.stft(). If window_length is one, then the returned window is a single element tensor containing a one.

Function Signature

```
torch.kaiser_window(window_length, periodic=True, beta=12.0,
*, dtype=None, layout=torch.strided, device=None,
requires_grad=False) → Tensor#

Parameters

Keyword Arguments
```

Parameters

Keyword

dtype (torch.dtype, optional) – the desired data type of returned tensor. Default: if None, uses a global default (see torch.set_default_dtype()). layout (torch.layout, optional) – the desired layout of returned window tensor. Only torch.strided (dense layout) is supported. device (torch.device, optional) – the desired device of returned tensor. Default: if None, uses the current device for the default tensor type (see torch.set_default_device()). device will be the CPU for CPU tensor types and the current CUDA device for CUDA tensor types. requires_grad (bool, optional) – If autograd should record operations on the returned tensor. Default: False.

Notes & Warnings

NOTE Note If window_length is one, then the returned window is a single element tensor containing a one.

Detailed Documentation

Documentation

Computes the Kaiser window with window length window_length and shape parameter

beta.

Let I_0 be the zeroth order modified Bessel function of the first kind (see torch.i0()) and N = L - 1 if periodic is False and L if periodic is True, where L is the window_length. This function computes:

Calling torch.kaiser_window(L, B, periodic=True) is equivalent to calling torch.kaiser_window(L + 1, B, periodic=False)[:-1]). The periodic argument is intended as a helpful shorthand to produce a periodic window as input to functions like torch.stft().

If window_length is one, then the returned window is a single element tensor containing a one.

window_length (int) – length of the window.

periodic (bool, optional) – If True, returns a periodic window suitable for use in spectral analysis. If False, returns a symmetric window suitable for use in filter design.

beta (float, optional) – shape parameter for the window.

dtype (torch.dtype, optional) – the desired data type of returned tensor. Default: if None, uses a global default (see torch.set_default_dtype()).

layout (torch.layout, optional) – the desired layout of returned window tensor. Only torch.strided (dense layout) is supported.

device (torch.device, optional) – the desired device of returned tensor. Default: if None, uses the current device for the default tensor type (see torch.set_default_device()). device will be the CPU for CPU tensor types and the current CUDA device for CUDA tensor types.

requires_grad (bool, optional) – If autograd should record operations on the returned tensor. Default: False.
